

Engineering Assessment

Technical assessment

Thank you for taking the time to complete our technical assessment for the position of Principal or Senior Data Engineer.

The assessment will ask you to accomplish a series of tasks that will end with a "production" ready extraction pipeline. While we would like you to accomplish every task, we understand that you may have other priorities or time constraints, so we encourage you to send what you have when you are ready. If you were limited on time, please ensure that you've documented what you would do for the subsequent tasks. However, what you do submit should be production ready.

We have tried to make this representative of the things that you may be asked to do at PFP. If you need any clarity, or get stuck, feel free to reach out to darren.abraham@placesforpeople.co.uk or andrew.farr@placesforpeople.co.uk.

The following task will have you:

- Create a simple ETL pipeline.
- Containerize the pipeline
- Create automated deployment

The dataset

- Use the Ducks Unlimited API to search for university chapters in the states of California, Oregon, Washington.
 - URL: [DU University Chapters](#)

Tasks

1. Write a simple ETL pipeline, in Python, that queries the Ducks Unlimited API's university chapter.
 - Return the chapters that are in the state of California (CA).
 - Bring in the:
 - Chapter Id
 - Chapter Name
 - City
 - State
 - Coordinates
2. Store the results in a Postgres table.
 - Use the Postgres database container.
3. Containerise the service using docker, so that it can be deployed and ran regularly.
 - Include a docker compose for local testing.

4. Write an IAC configuration to deploy the service to a cloud service of your choice.
 - Build and deploy the image to a container registry. If this is done in a private registry ensure that it is defined in the IAC.
 - You can work under the assumption that the service is accessing a cloud hosted Postgres instance, or deploy the cloud hosted instance.
 - Any schedule should be specified in the IAC, run this service daily.
5. Create a CICD pipeline to deploy the image.

Notes

- What we're looking for
 - Quality, clarity of intent, readability
 - A well structure codebase with appropriate tooling.
 - Good separation of concerns
 - Sensible file structure
 - Testing
 - Error handling
 - Informative documentation
 - setup
 - usage
 - architecture.
 - Reproducibility
 - Ensure that this can be ran locally by a member of our team
 - You will also be asked to walk through your code and run it for us.
- Manage your time effectively
 - Prefer an end to end process over perfect code.
- Deliver your codebase in a Git repo (Github or Gitlab) then grant access to the following developers:
 - Github: Andrew-data-eng
 - Gitlab: Andrew-data-eng
- Use whatever tools you would like to accomplish these tasks.
 - Avoid having Claude Code write the entire thing, the work should be yours, you will be asked to walk through and justify decisions made.

Please monitor your cloud spend and usage! Considering you will likely be testing this with a free tier from your cloud provider of choice, we do not want you incurring any unexpected costs.